Gender differences among popular musicians on YouTube and Spotify

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https://github.com/caragrimm/SMDA-Project

Abstract

This study explores gender differences among popular musicians on YouTube and Spotify, focusing on popularity metrics and sentiment in comments. Data from the recent top 20 global artists revealed no significant gender-based differences in popularity metrics such as views, likes, and followers. However, sentiment analysis of YouTube comments showed that female musicians receive a higher proportion of negative comments compared to their male counterparts. These findings highlight a gender bias in online feedback despite similar overall popularity, suggesting the need for continued efforts to address gender disparities in digital music platforms

1 Motivation

The music industry has long mirrored broader societal norms and biases, including gender imbalances. Female musicians, along with other underrepresented groups, have historically encountered considerable barriers to success. These barriers include a lack of connections and female role models in the industry, uncomfortable studio environments, sexualization, and frequent underestimation and stereotyping (Epps-Darling et al. 2020, p. 248). These disparities are also evident in music charts and award nominations; for instance, women constituted only 20.2% of the artists on the 2020 Billboard Hot 100 year-end chart and just 13.4% of Grammy nominees from 2013 to 2021 (Smith et al. 2021). This project seeks to explore whether these gender disparities among musicians extend to online music consumption. With YouTube and Spotify being among the leading audio streaming platforms in the US and UK (Statista 2022), this study will analyze whether there are significant differences in popularity metrics between male and female musicians on these platforms. Additionally, it will investigate whether the overall sentiment of comments on YouTube differs by the musician's gender. Previous studies have indicated that female creators tend to receive a higher volume of negative comments compared to their male counterparts (Bertaglia et al. 2023; Amarasekara and Grant 2019). Understanding these differences is crucial for highlighting potential biases and trends in online music consumption and gaining insights into how gender influences visibility and engagement in the digital music landscape.
The project addresses the following two research questions:

1. Does being a female musician result in significantly lower popularity metrics (views, likes, comments and subscriber counts on YouTube and followers and popularity scores on Spotify) compared to male musicians?

2. Does being a female musician result in a more negative overall sentiment of YouTube comments compared to male musicians?

2 Data retrieval

To address the research questions, data was collected from the YouTube Music Charts, YouTube API, Spotify Charts, and Spotify API. The first step involved identifying the current top global artists. This was done by obtaining the top 100 artists from the YouTube Music Charts (Global Weekly Top Artists) and Spotify Charts (Weekly Top Artists Global). The YouTube Music Charts data is available for download as a CSV file and reflects the total view count of an artist's entire discography (Google 2024). For Spotify Charts, data was manually copied due to restrictions on web scraping. These charts are based on a specific non-public formula from Spotify (Spotify 2024c). The top 100 artists from both platforms were retrieved over a four-week period (June 28, 2024, to July 25, 2024), allowing for the identification of the recent top 20 global artists.

2.1 Spotify API

An app was created on the Spotify Developer Website to obtain credentials and interact with the API. Artist IDs for the top 20 artists were retrieved, allowing for the collection of each artist's follower count and popularity score. The popularity score, ranging from 0 to 100, reflects the artist's overall popularity based on their tracks (Spotify 2024b).

2.2 YouTube API

For YouTube, the API client was initialized using a generated API key. Initially, YouTube channel IDs for the top 20 artists were retrieved. Due to some discrepancies with similar artist names, manual corrections were made to ensure accurate channel statistics. Channel statistics, including view counts, subscriber counts, and video counts, were collected from the channel endpoint. Furthermore, data for all videos from each channel, such as views, likes, and comments, was collected from the video endpoint and aggregated for each artist. A discrepancy was noted between the number of videos reported by the channel endpoint and the video endpoint. YouTube does not clearly define the composition of video counts on the channel endpoint, the discrepancy may be attributed to private or unlisted videos. For further analysis, the number of videos obtained from the video endpoint was used to ensure consistency. To manage API quota limitations, the comment retrieval for the second research question was segmented. The 20 most recent videos per channel were retrieved, focusing exclusively on top-level comments and excluding replies. Channels were organized into four groups, with comments collected and saved separately for each group. These separate comment sets were then combined for a comprehensive analysis.

3 Data processing

The collected data underwent several processing steps to ensure accuracy and consistency. Initially, weekly top 100 artist data from YouTube and Spotify were loaded into separate DataFrames. These DataFrames were merged sequentially to create a single DataFrame representing the top artists over the four-week period. The YouTube and Spotify data were then combined on artist names to create a comprehensive view of artist rankings across both platforms. Bands were excluded from the analysis, as the focus was solely on solo musicians. The resulting dataset was saved to a CSV file, and each artist's gender was manually labeled using online sources. To ensure a balanced top 20 list, the ranks from both the Spotify and YouTube charts were summed for each artist. The DataFrame was sorted by the combined rank, and the top 10 male and top 10 female artists were selected to form the top 20 recent global musicians for further analysis.

Once the popularity metrics were retrieved, the average view count, like count, and comment count per video were calculated for each artist. To address skewed distributions and outliers in the YouTube metrics, log transformations were applied to these metrics, normalizing the data and enhancing the robustness of statistical analyses. The data was then aggregated by gender, calculating various metrics such as total and average subscribers, followers and channel views, as well as average popularity scores, and average log-transformed video statistics.
Before performing sentiment analysis for the second research question, several processing steps were performed to ensure uniformity. Comments were converted to strings, HTML tags were removed, URLs were eliminated, and all text was converted to lowercase. The text was then tokenized into individual words and reassembled into cleaned strings. Only English comments were retained for analysis using VADER and BERT, as these tools are specifically designed and trained for English language sentiment analysis. Language detection was performed using the `langdetect` package from PyPi. Videos without English comments were excluded, resulting in 788,281 English comments. The comments were then parsed from string formats into lists, grouped by artist and video, and categorized by gender. To ensure balanced representation, a sample of 100,000 comments was drawn for each gender, including a proportionate number of comments from each artist.

4 Analysis

4.1 Popularity Metrics

The analysis of popularity metrics revealed variability across the metrics, with some artists exhibiting exceptionally high values and others much lower. This variation may be attributed to factors such as the relative novelty of some artists or their association with niche genres. Visualizations, including boxplots and histograms, highlighted significant outliers in metrics such as average view count, like count, and comment count per video. These visualizations also confirmed that the data is not normally distributed. Given these characteristics and the small sample size, non-parametric statistical tests were employed. The Mann-Whitney U test and permutation tests were used to evaluate gender-based differences in popularity metrics. The data met the assumptions of the tests of having independent groups, similar distribution shapes across groups, and continuous values. The Mann-Whitney U test assesses whether there is a significant difference in the distribution of a metric between male and female artists by ranking the data and comparing the rank sums of the two groups (Eid et al. 2017, pp. 343; Holt and Sullivan 2023, pp. 779; Emerson 2023, pp. 99). To complement this, permutation tests were also conducted, which involve repeatedly shuffling the data and recalculating the test statistic to assess the likelihood of observing the results under the null hypothesis (Holt and Sullivan 2023, pp. 779). In contrast to the initial research question one, Figure 1 shows that female artists have higher total and average numbers of YouTube subscribers and channel views than their male counterparts. Additionally, when examining logarithmic averages for views, likes, and comments per video, the data suggests that, after adjusting for outliers, male and female artists have similar average engagement metrics. On Spotify, while the average popularity appears comparable between genders, female artists tend to have higher total and average numbers of followers. Despite these visual observations, the statistical analyses revealed no significant differences in popularity metrics between female and male musicians. Both the Mann-Whitney U test and permutation tests resulted in p-values greater than 0.05 for all examined metrics, indicating that the differences seen in the visualizations are not statistically significant. Therefore, the analysis confirms that gender does not lead to significantly lower popularity metrics for female musicians compared to their male counterparts on YouTube and Spotify. This suggests that, contrary to the initial hypothesis, gender does not significantly affect popularity metrics in this dataset.

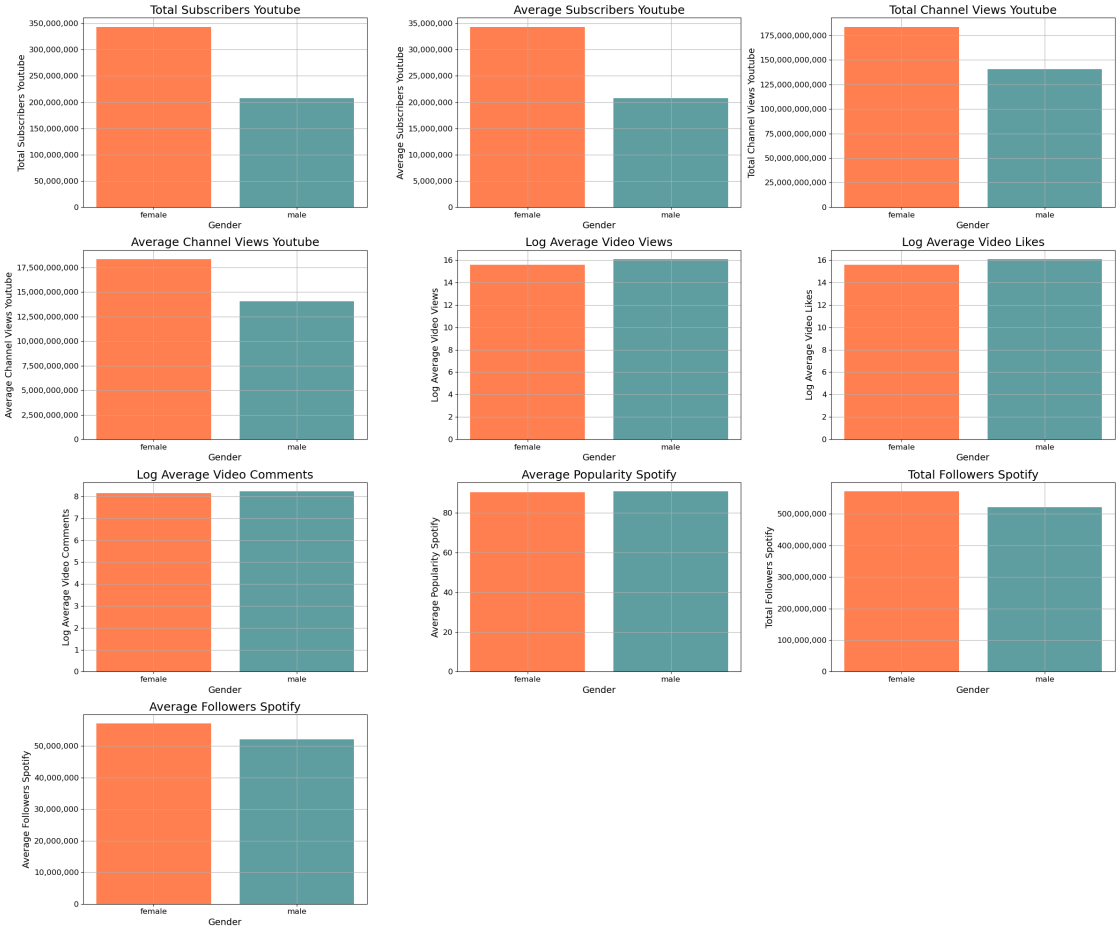

Figure 1: Popularity Metrics of YouTube and Spotify by Gender

4.2 Sentiment Analysis

To investigate whether female musicians receive a higher proportion of negative comments compared to male musicians, sentiment analysis was performed on a sample of 100,000 comments using VADER and BERT. The BERT model employed was the `twitter-roberta-base-sentiment-latest`, pretrained on 124 million tweets. Both tools classified comments as positive, negative, or neutral. While the BERT model directly provides these labels, the VADER's sentiment labels are based on classification of the compound scores recommended by the VADER documentation (positive: (compound score 0.5); neutral:(-0.5 < compound score < 0.5); and negative: (compound score -0.5)) (Hutto 2021). A two-proportion z-test was employed to compare the proportion of negative comments between genders, chosen for its suitability in comparing proportions between two independent groups and large sample sizes (Nelson 2023, p. 103). The results revealed a statistically significant difference, with female musicians receiving a higher proportion of negative comments than their male counterparts, as indicated by p-values of 0.0000 for both VADER and BERT. Additionally, chi-square test of independence was conducted to examine the relationship between gender and sentiment categories. This test determines whether there is a significant association between two categorical variables (Nelson 2023, p. 105). The chi-square test also showed a significant relationship, confirming that sentiment varies significantly by gender. Figures 2 and 3 present visualizations of the sentiment proportions. These figures illustrate that female musicians

receive a larger share of negative comments, with BERT detecting a more pronounced difference compared to VADER. Figure 3 further shows that female musicians receive fewer neutral comments overall, and the results for positive comments varied between the two tools. In summary, both statistical tests and sentiment analysis tools confirm that female musicians experience a more negative overall sentiment in YouTube comments compared to male musicians.

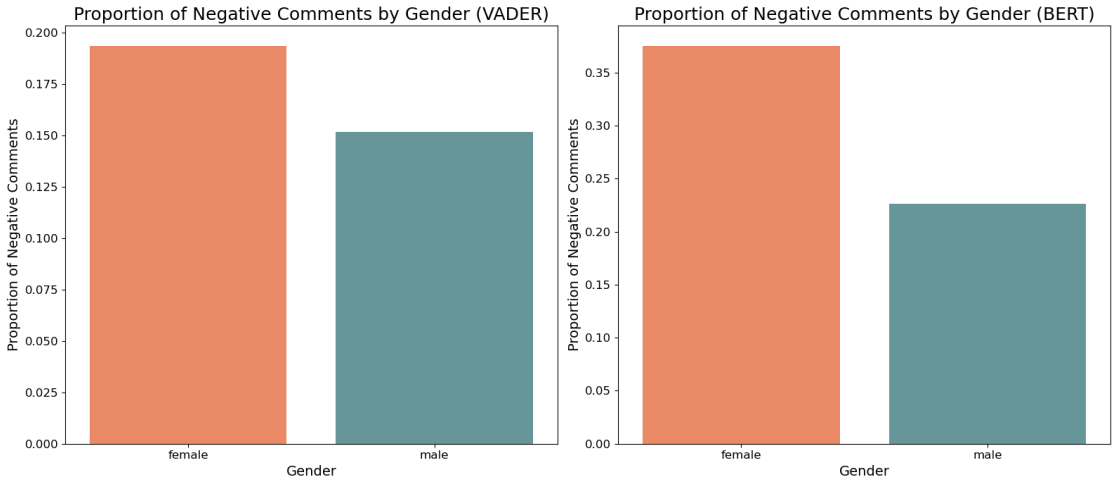

Figure 2: Proportion of Negative Comments by Gender Using VADER and BERT

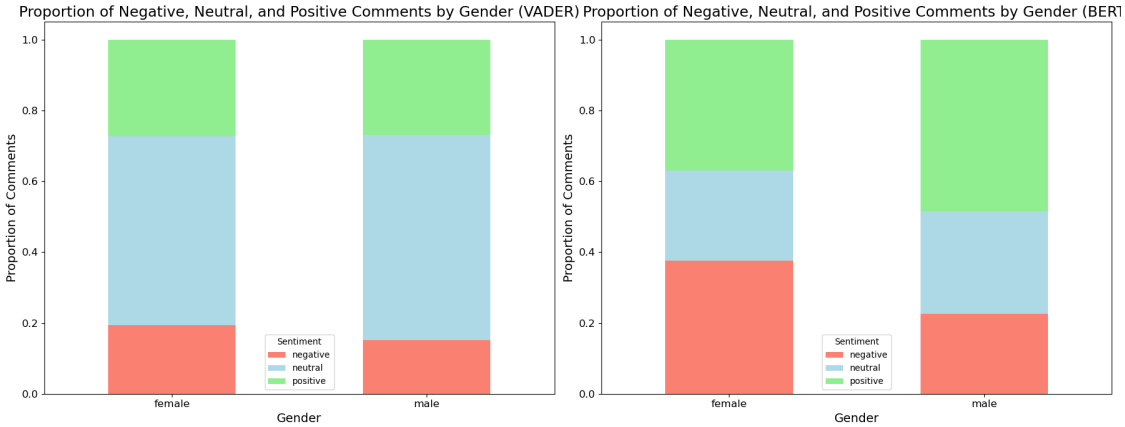

Figure 3: Distribution of Negative, Neutral, and Positive Comments by Gender Using VADER and BERT

5 Conclusion

This project provides nuanced insights into gender differences among popular musicians on YouTube and Spotify. In terms of popularity metrics, the analysis found no significant differences between male and female musicians. Despite visual evidence suggesting higher engagement metrics for female artists on YouTube and Spotify, statistical tests, including the Mann-Whitney U test and permutation tests, did not find these differences to be statistically significant. This suggests that, based on the metrics examined, gender does not appear to have a significant impact on the overall

popularity on these platforms. However, a different pattern emerges when looking at sentiment in YouTube comments. Sentiment analysis using both VADER and BERT revealed a significant difference in the type of comments received by male and female musicians. Female artists received a higher proportion of negative comments than their male counterparts, as confirmed by the two-sample z-test and chi-square tests of independence. This finding highlights a gender bias in online comments, with female musicians facing more negativity, which is consistent with previous research. These findings highlight an important dimension of gender inequality in the music industry: while overall popularity metrics may not reflect a significant gender gap, the sentiment expressed in online comments reveals a clear bias. This highlights the need for efforts to address gender inequality and biases within digital environments, ensuring that all musicians, regardless of gender, receive fair and equitable treatment.

6 Critique

This project offers a valuable examination of gender differences among musicians on YouTube and Spotify, yet several limitations must be acknowledged. One significant limitation is the potential inaccuracy in gender labeling, particularly when self-identification is not available. Gender is not always static and may evolve over time (Epps-Darling et al. 2020, p. 253). Additionally, the study focuses solely on binary gender categories, excluding gender minorities such as non-binary artists. This exclusion limits the inclusivity of the research (Bertaglia et al. 2023, p. 27; Wang and Horvát 2019, p. 519). Another major limitation is the narrow scope of the study, which examines a specific subset of the top 20 global solo artists over a four-week period. This limited focus may fail to capture broader trends across different time periods, genres, or types of artists. The small sample size could also undermine the robustness of the conclusions. Future research with a larger and more diverse sample could offer more definitive insights and potentially uncover gender-related disparities that the current study could not detect. The study also faces constraints related to data completeness. API limitations, such as rate limits and data availability, affect the extent of the data collected. While metrics like followers, views, likes, and comments are useful for quantifying popularity, they may not fully reflect an artist's success or engagement. Observed variability and outliers in the data might be influenced by factors such as viral content or recent releases, which can skew the results. Sentiment analysis presents another challenge. The choice to sample 100,000 comments per gender might not fully represent the variety of comments. In addition, the reliance on sentiment analysis tools like VADER and BERT, while effective, has limitations, including difficulty capturing nuances like sarcasm, humor, and ambiguous language, which may lead to misclassification of comments. The differences in sentiment classification between these tools highlight the complexity of sentiment analysis and suggest that results may vary depending on the tool used. Moreover, attempts to filter for English comments were not entirely successful, as non-English comments were still present in the dataset. This could introduce inaccuracies into the sentiment analysis, potentially affecting the project's conclusions. Furthermore, relying on English comments might not accurately represent global sentiment, and the comments may not reflect the views of all viewers, as they capture a more vocal subset of the audience. It is also important to mention that the project does not account for intersectional factors like race, age, or genre, which could influence the popularity metrics and sentiment. Temporal dynamics also play a role, as sentiment and popularity metrics can change over time, potentially impacting the findings. In summary, while the project makes a significant contribution to understanding gender dynamics in online music platforms, addressing these limitations could improve the depth and accuracy of the analysis and provide a more nuanced view of gender inequalities in the music industry.

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